

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Athavan K	Reg. No.:	192524036
Section / Batch:	AI&DS	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1

SECTION A – CONCEPT MAPPING

Q1) Explain how nested loops increase computational cost. Extend the map with examples such as $O(n^2)$ and $O(n^3)$. Justify why higher polynomial complexity affects scalability. Interpret how this mapping identifies inefficient designs

Code of Conduct

I Athavan K(192524036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Athavan K

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%				
Linkages	25%				
Organization	20%				
Integration of Literature	20%				
Clarity	10%				
Total Marks (100):					

COL - AT - 2

Concept Mapping

How Nested Loops Increase Computational Cost

Nested loops repeat operations multiple times. Each inner loop multiplies the number of operations of outer loop.

1. How Nested Loops Increase Cost

- Outer loop of size n runs n times
- Inner loop of size n runs n times each outer iteration
- Total operations = $n \times n = n^2$
- Adding more nesting levels multiplies the repetition further.

2. Extend the Map with Examples

A) $O(n^2)$ - Two Nested Loops

Structure:
for $i = 1$ to n
 for $j = 1$ to n
 do constant work

Total operations: $n \times n = n^2$

Examples:

- Bubble Sort / Selection Sort
- Matrix Traversal
- checking all pairs in array

B) $O(n^3)$ - Three Nested Loops

Structure:
for $i = 1$ to n
 for $j = 1$ to n
 for $k = 1$ to n
 do constant work

Total operation: $n \times n \times n = n^3$

Examples:

- Matrix Multiplication
- Floyd-Warshall Algorithm
- 3D array traversal

Name: Atharva K

Reg No: 192524036

Course code: CSA0614

Course: Design & Analysis of Algorithms

3. Why higher Polynomial Affects Scalability

- As n grows, operations grow very fast.
- $O(n^2)$ grows quadratically - 10x input means 100x operations
- Leads to higher time, memory and resource usage.